

6 A Clockwise moment by weight of M = anticlockwise moment by tension in string

$$Mg(1.200) = 300 \sin 30^\circ (0.805)$$

$$M = 10.3 \text{ kg}$$

7 C Net force

$$= \text{Upthrust by air} - \text{Weight of balloon} - \text{Weight of balloon skin} - \text{instrument}$$